

Assignment 1: LLM-Powered Chatbot System

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GitHub Link : [Chatbot 1441 1445](#)

2. Project Overview:

This project implements a lightweight LLM-powered chatbot with user authentication, real-time AI responses, and persistent chat history. The system uses a locally-hosted open-source model (Qwen 1.8B via Ollama), ensuring privacy and offline functionality. The full-stack application features a FastAPI backend with SQLite and a React frontend.

3. System Architecture:

Backend: FastAPI (Python), SQLite via SQLAlchemy, JWT authentication, Ollama API integration

Frontend: React.js + Vite

AI Model: Qwen 1.8B — lightweight open-source LLM running locally via Ollama inference server

4. Key Features:

- Secure login & registration with JWT token authentication
 - Persistent chat history — all conversations saved per user account
 - Multiple chat sessions with rename and delete support
 - Fully local AI inference — no external API calls required
 - Animated AI face with idle, thinking, talking, and happy expression states
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5. How to Run

- Install Ollama and pull model: `ollama pull qwen:1.8b`
 - Start backend: `uvicorn app.main:app --reload --port 8000`
 - Start frontend: `npm run dev` → open `http://localhost:5173`
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6. Conclusion

This project successfully demonstrates integration of a locally-hosted LLM with a full-stack web application, satisfying all requirements: user authentication, AI-powered conversation, and persistent chat history. The system highlights practical skills in API development, frontend design, and on-device AI model deployment.